

Module Introduction

CST3144 Full Stack Development

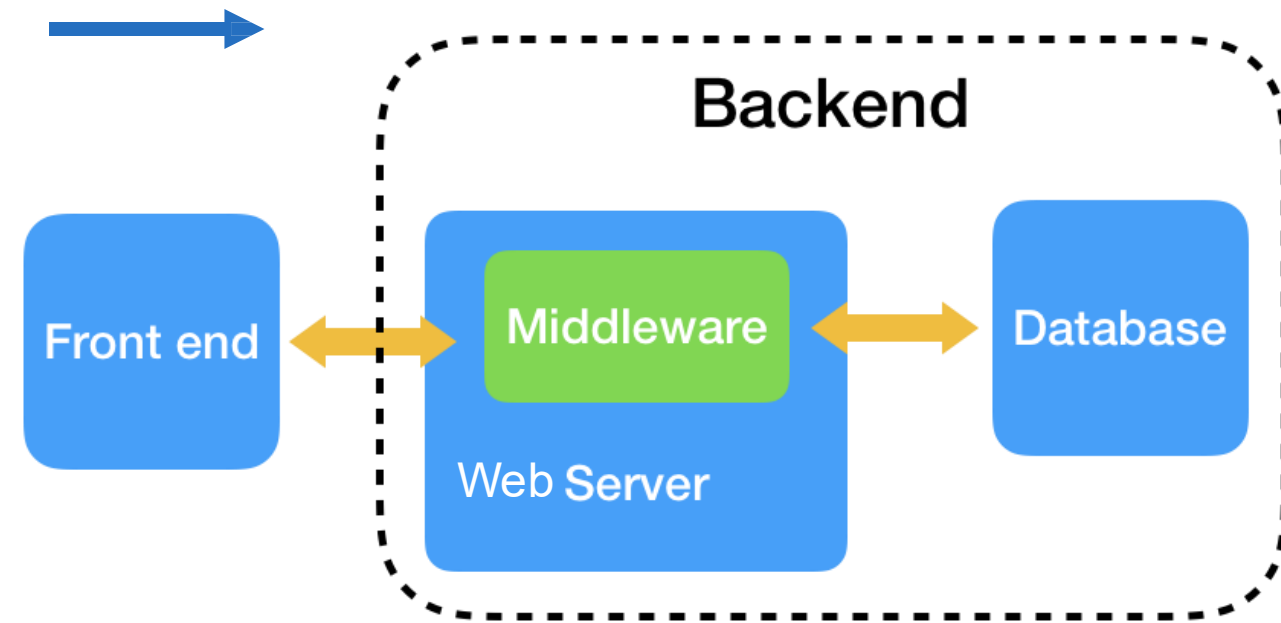
- **Created based on:**
 - the Students' feedback and
 - app development popularity
 - CST3145 Web-Based Mobile App Development
- The Module will cover the **3 most important components** of a complete architecture for a software application:
 - **Front-End, Back-End, and System Administration**
- **The details of the contents and assessments** can still **change** over the year:
 - there could be slight changes depending on **(Module) External Examiner**
 - **minor changes/updates** to the content
 - **material released** usually **week by week** after my further/final **revision**

Intro: Full Stack Development

What is a Web Development Stack?

- All the components needed to run a Web app: →

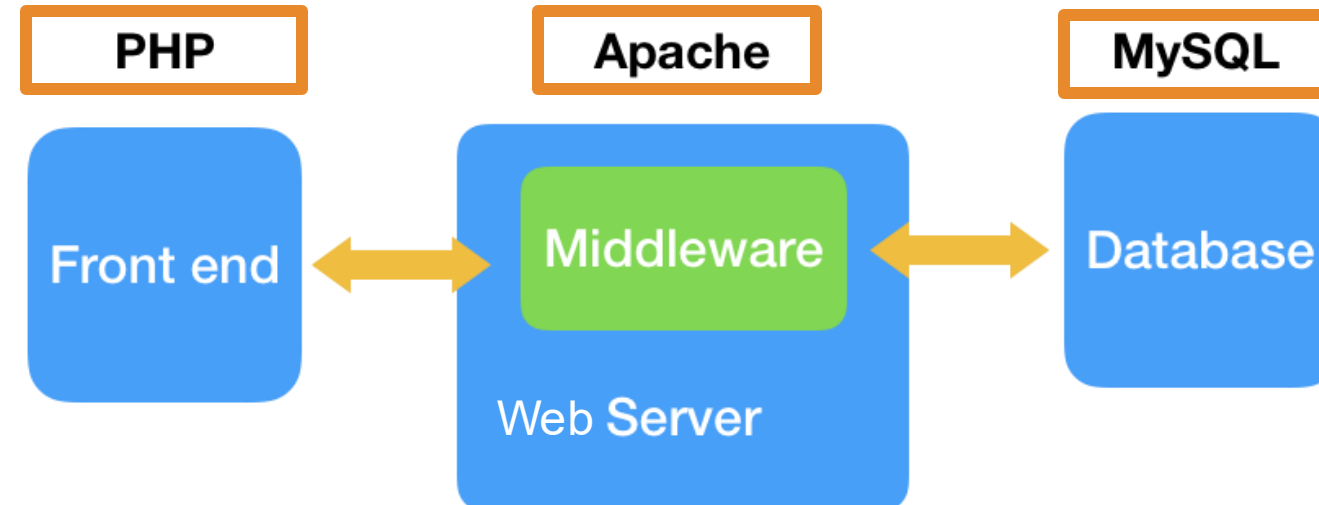
- Front-End
- Back-End: Web Server, Middleware, Database



- Popular stacks, for instance **xAMP** ↘

- **xAMP**: **x**, Apache, MySQL and PHP
- **LAMP**: Linux, Apache, MySQL and PHP
- **MAMP**: MacOS, Apache, MySQL and PHP
- **WAMP**: Windows, Apache, MySQL and PHP

- Multiple languages: PHP, Java and SQL



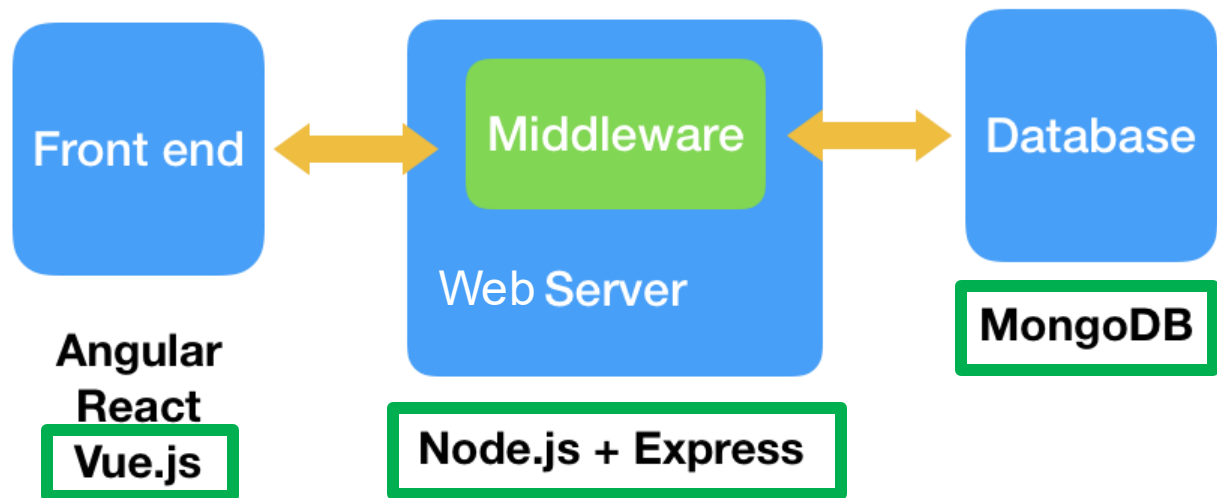
Popular Stacks MExN and Our Approach

Everything
will be
hosted
online

Other Popular stacks, MExN

- **MExN:** MongoDB, Express, **x** and Node.js
- **MEAN:** MongoDB, Express, **Angular** and Node.js
- **MERN:** MongoDB, Express, **React** and Node.js
- **MEVN:** MongoDB, Express, **Vue.js** and Node.js

- **[Advantage]** not using multiple languages,
 - **only using JavaScript** (MongoDB is “JavaScript-Friendly” by using JSON like structure)

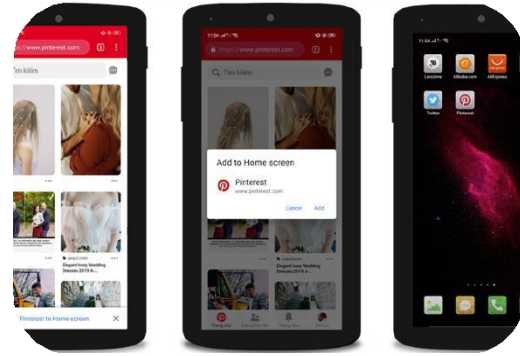
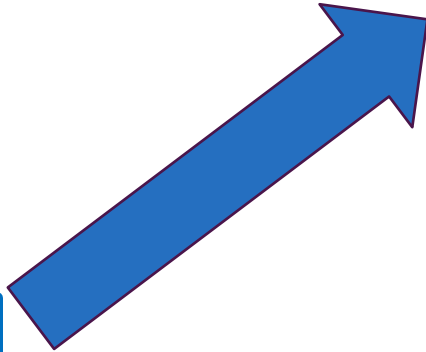


Our Approach is Project-Based and:

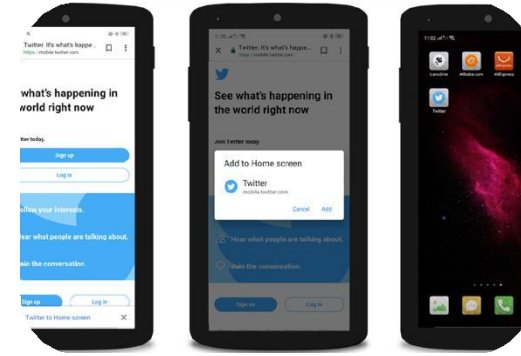
- We will focus on **MEVN:** MongoDB, Express, **Vue.js**, and **Node.js**
- **We cover Vue.js as first framework**
- Then we cover **Node.js (the server)**
- Then **Express.js (middleware)**, which is built on top of Node
- We will briefly cover **MongoDB (database)**, as **it is covered in the 2nd-year web dev module**
- We will also cover **how to host Node/Express** (within **Amazon Web Services, AWS**) and **MongoDB (Atlas) online**.

Full-Fledged Application

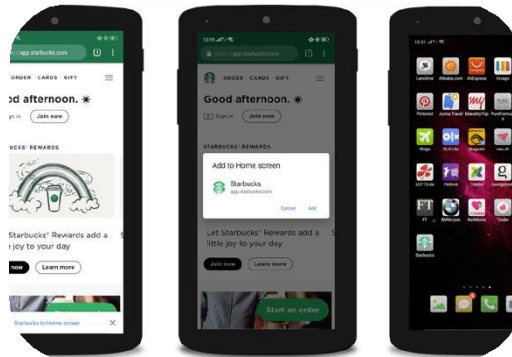
- Full Stack Development is:
 - mature** and **well-established**
 - widely used**
- Many of the **most used apps** follow such approach:
 - Facebook
 - Instagram
 - AirBnB
 - Uber
 - Discord
 - Skype
 - Pinterest
 - Twitter Lite
 - Starbucks
 - ...



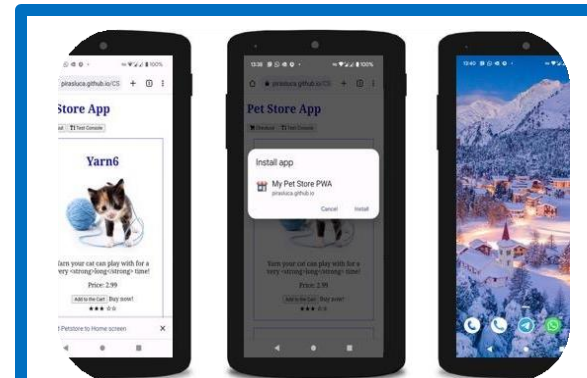
Pinterest



X (Previously Twitter)



Starbucks



Our Final App
(Project-Based Approach)

Teams Link to Join the Module



Suggestions for Reading

Reading

- Module Handbook

Questions?